

EXPERIMENTAL PROTOCOL

Completion of the Performance Assessment of the Unitree L1 Three- Dimensional LiDAR

*Controlled range metrology, non-repetitive coverage, Robot Operating System 2 stream stability,
and trajectory validation*

PURPOSE: This protocol defines the experiments required to close the principal evidence gaps identified in the preceding evidence-based assessment. It is designed for direct execution in the Human-Centered Mobile Robotics Laboratory and for subsequent use in a short scientific paper.

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SCIENTIFIC WRITING CONVENTION: All acronyms are expanded at first use. Manufacturer specifications, direct observations, derived quantities and scientific inferences are kept distinct. Tables are limited to decisions, test matrices and metadata that cannot be read more clearly in prose.

1. Purpose, research question and evidential scope

The Unitree L1 is a compact light detection and ranging sensor (LiDAR) with a hemispherical, non-repetitive scan pattern. Published evidence supports navigation-scale mapping under favourable conditions, but four questions remain unresolved: controlled range accuracy, time-dependent angular coverage, long-duration Robot Operating System 2 (ROS 2) stability, and trajectory accuracy against an independent reference. These gaps are identified in the preceding evidence assessment [1].

The research question is: under which controlled operating and processing conditions does the Unitree L1 provide measurements, spatial coverage and data streams that are sufficiently stable for indoor robotic perception and mapping?

The protocol separates four levels of performance. Range metrology concerns the estimated distance to a known surface. Angular coverage concerns the distribution of valid returns over space and time. Stream stability concerns message timing and data integrity. Trajectory validation concerns the combined sensor-processing system. A result at one level must not be presented as evidence at another level.

1.1 Hypotheses

- H1 - Controlled range performance. On a matte, frontal target and within the declared operating range, the mean signed range error is expected to be compatible with the manufacturer-stated ± 0.02 m accuracy under comparable conditions. Dispersion and invalid returns are expected to increase with range and incidence angle.
- H2 - Non-repetitive coverage. The fraction of observed angular cells increases with accumulation time and is direction dependent. Consequently, a single nominal point rate is insufficient to describe usable spatial density.
- H3 - Stream stability. The point-cloud and inertial streams can operate continuously without timestamp reversal or driver restart, but their effective rate, timing jitter and invalid-return fraction must be measured rather than inferred from the manual.
- H4 - System-level validation. With a fixed LiDAR-inertial processing pipeline, repeated indoor trajectories produce bounded trajectory and map disagreement. The magnitude of this disagreement depends on speed, geometric overlap and the quality of the independent reference.

1.2 Scope and non-claims

Experiments 1 to 3 isolate sensor and software behaviour. Experiment 4 evaluates the complete odometry and mapping system. The protocol does not treat an Ouster-64 map or a LiDAR-inertial algorithm as absolute metrological truth unless that reference has itself been validated against motion capture, a total station or an equivalent traceable system. If no independent reference is available, Experiment 4 is reported as repeatability only.

1.3 Required experimental blocks

Block	Primary outcome	Status	Minimum repetition
1. Planar target	Bias, repeatability, point-to-plane dispersion, valid-return rate	Mandatory	5 independent acquisitions per condition
2. Static coverage	Coverage-versus-time, directional density, revisit behaviour	Mandatory	3 independent 60 s acquisitions
3. ROS 2 stability	Rates, jitter, gaps, invalid points, resource use	Mandatory	1 continuous 60 min run; 30 min minimum
4. Trajectory validation	Trajectory error and map consistency against a reference hierarchy	Mandatory for accuracy claims	5 traversals per tested speed; 3 minimum

2. Experimental design and general controls

The acquisition, processing and exclusion rules are fixed before the definitive runs. Each independent acquisition is treated as one statistical unit. Individual points within one point cloud are correlated and must not be counted as independent experimental replicates.

2.1 Hardware identification and configuration freeze

1. Photograph the sensor label and record the exact L1 variant, serial number, firmware version and physical orientation. The Precision Measurement and Remote Measurement variants have different declared range limits [2].

2. Record the computer model, central processing unit, memory, network interface, power supply and mounting arrangement.
3. Freeze the driver, ROS 2 distribution, container image, launch files, parameter files, coordinate transforms and analysis commit. Record immutable identifiers such as Git hashes and container image digests.
4. Clean the optical window. Warm the sensor for 10 min before definitive measurements. Keep the same warm-up procedure for all runs.
5. Use one coordinate convention throughout. Record the location of the LiDAR origin and the transformation between the sensor frame, robot frame and reference system.

2.2 Environmental controls

Record ambient temperature, approximate illumination, direct sunlight, air movement, nearby reflective surfaces and human motion. Experiments 1 to 3 are performed in a static indoor environment. Doors remain closed when practical. No person stands within the measurement region during acquisition. Any unavoidable environmental change is recorded as a protocol deviation.

2.3 Repetition, order and exclusion

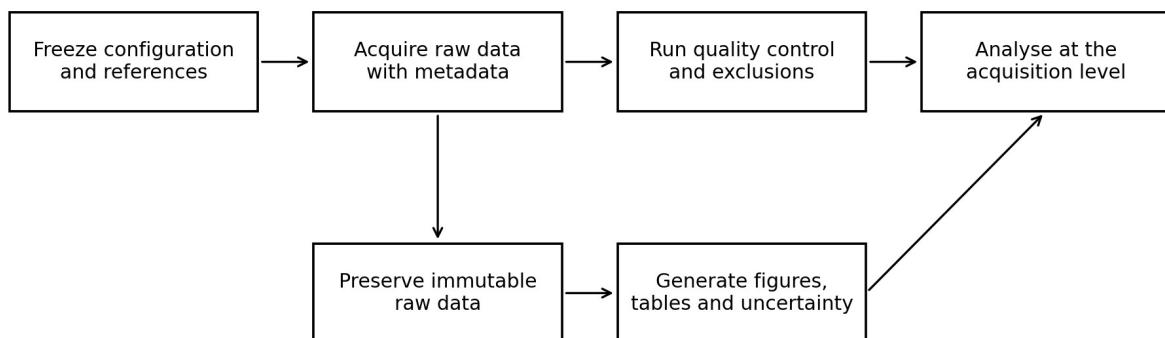
Five independent acquisitions are used for each controlled range condition. The order of distances is randomized within each block when relocation time permits. Angle and material conditions are also randomized. This design reduces confounding by warm-up or slow thermal drift. A run is excluded only under a pre-defined condition: incomplete recording, missing required topic, physical movement of the target or sensor, persistent frame error, timestamp reversal, or an undocumented configuration change. Excluded runs are retained and labelled; they are not deleted.

2.4 Reference uncertainty

The reference distance is measured with a calibrated laser distance meter, total station or verified steel tape. The resolution and calibration status are recorded. The geometric offset between the measured surface and the LiDAR coordinate origin is surveyed. The combined standard uncertainty is estimated from independent components by root-sum-square combination:

$$u_c = \sqrt{u_{\text{reference}}^2 + u_{\text{origin}}^2 + u_{\text{alignment}}^2 + u_{\text{positioning}}^2}$$

A comparison with a ± 0.02 m manufacturer specification is meaningful only when the expanded reference uncertainty is substantially smaller than 0.02 m. If this condition is not met, the result is reported as an observed difference under the available reference accuracy, not as verification or rejection of the specification.



All processing parameters, software versions and random seeds are version controlled.

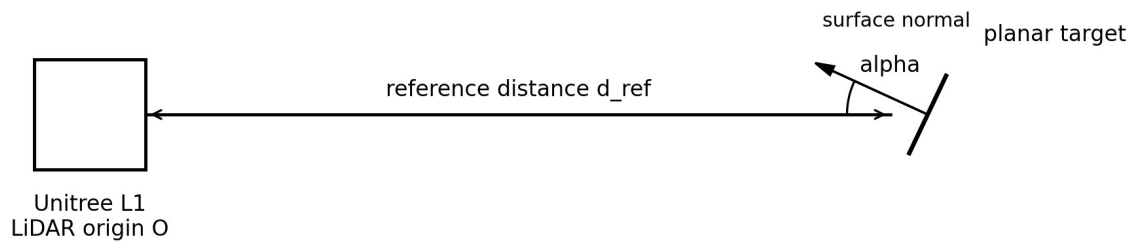
Figure 1. Experimental workflow. Raw recordings remain immutable; quality control and all derived results are generated by version-controlled scripts.

3. Experiment 1 - controlled planar-target range performance

This experiment measures systematic range error and short-term repeatability under controlled geometry. It does not estimate complete sensor uncertainty over every surface and environmental condition.

3.1 Equipment and geometry

- A flat target at least 1.0 m by 1.0 m. A rigid matte white panel is the primary target. A matte black panel of comparable flatness is used for the material comparison.
- A calibrated distance reference with millimetre-level resolution when possible.
- A rigid sensor mount and an adjustable target support with an angle scale or digital inclinometer.
- Floor markers or a linear rail to reproduce sensor-target positions.
- A static room with sufficient clearance for the maximum tested distance.



Measure d_{ref} from the LiDAR coordinate origin, or correct for the surveyed origin offset.

Figure 2. Planar-target geometry. The incidence angle α is measured between the incoming line of sight and the target surface normal.

3.2 Test matrix

Series	Controlled factors	Levels	Independent runs	Duration per run
Distance block	Matte white, $\alpha \leq 3$ degrees	0.5, 1, 2, 3, 5, and 8 m; add 10 m if space permits	5	20 s
Incidence block	Matte white, fixed distance 2 or 3 m	0, 30, and 60 degrees	5	20 s
Material block	Fixed distance 2 m, $\alpha \leq 3$ degrees	Matte white and matte black	5	20 s
Exploratory surfaces	Fixed distance 2 m	Metal or glass only if available	3	20 s

The exploratory surfaces are not pooled with matte targets because specular and transmissive behaviour differs qualitatively. Their results are reported as observations, not as a calibrated reflectance study.

3.3 Acquisition procedure

1. Position the target so that its centre lies within the dense central field of view. Confirm that the target fills the selected region of interest and that no edge intersects that region.
2. Measure the reference distance from the defined LiDAR origin to the target plane. Repeat the measurement three times. If the spread exceeds 0.003 m, inspect the alignment and repeat.
3. Measure the target angle. Record both the nominal angle and the instrument reading.
4. Wait 5 s after any physical adjustment. Confirm that the monitoring node reports valid point-cloud and inertial messages.
5. Record 20 s of raw point-cloud and inertial measurement unit (IMU) data. Do not apply filtering during acquisition.
6. Reopen the bag immediately. Verify duration, topic presence, message counts and monotonic timestamps.
7. Complete the manifest before changing the setup. Record anomalies even when the run is accepted.

3.4 Processing and outcome definitions

Transform all points to the LiDAR frame. Select a central target region that excludes edges. Fit a plane by Random Sample Consensus (RANSAC) to reject outliers, then re-estimate the plane from inliers by total least

squares or singular value decomposition. The plane normal is oriented consistently before calculating the signed distance.

$$e_j = d_{\text{hat},j} - d_{\text{ref},j}$$

For acquisition j , $d_{\text{hat},j}$ is the estimated plane distance and $d_{\text{ref},j}$ is the corrected reference distance. The following acquisition-level outcomes are reported: signed error, absolute error, point-to-plane root-mean-square error (RMSE), median absolute deviation (MAD), valid-return fraction and number of inlier points. Condition-level results include the mean signed error, standard deviation across acquisitions and a 95% confidence interval.

PRIMARY DECISION RULE: The ± 0.02 m claim is evaluated only for the matte frontal condition that most closely matches the stated manufacturer test condition. The confidence interval and reference uncertainty are reported. No global accuracy claim is made from this single condition.

4. Experiment 2 - non-repetitive angular coverage

The L1 does not reproduce an identical angular pattern in every short interval. This experiment measures how valid spatial coverage develops with accumulation time. It characterizes the sensor in a defined indoor scene, not the theoretical path of every emitted ray.

4.1 Acquisition environment

Place the sensor motionless near the centre of a structured room containing walls, ceiling and objects at several ranges. Avoid moving people and large glass surfaces in the primary series. Record a panoramic photograph and a sketch of the sensor orientation. Acquire three independent 60 s sequences without changing the pose. A second orientation tilted by 30 degrees is optional but valuable for mounting guidance.

4.2 Angular representation

For every finite point (x, y, z) , calculate azimuth θ and elevation ϕ :

$$\theta = \text{atan2}(y, x), \quad \phi = \text{atan2}(z, \sqrt{x^2 + y^2})$$

Bin valid points in a 2 degree by 2 degree angular grid. Repeat the analysis with a 1 degree by 1 degree grid as a sensitivity check. Define the observable-cell set from the union of cells occupied during the complete 60 s reference acquisition. This definition prevents empty directions caused by room geometry from being counted as sensor failures.

4.3 Accumulation windows and metrics

Evaluate cumulative windows of 0.10, 0.25, 0.50, 1, 2, 5, 10, 20, 30 and 60 s. For each window, calculate:

- relative angular coverage $C(t)$: occupied observable cells at time t divided by observable cells in the 60 s reference;
- time to 50%, 90% and 95% of final coverage;
- hit-count coefficient of variation across occupied cells;
- median and 5th-to-95th percentile hit count by elevation band;
- revisit interval for cells receiving at least five returns;
- valid point count per message and per second.

Coverage curves are reported for each independent acquisition and as an acquisition-level mean with variability. Point counts from one run are not treated as independent replicates.

5. Experiment 3 - Robot Operating System 2 stream stability

This experiment tests continuous delivery and computational stability. It is separated from mapping so that driver or timestamp failures are not confused with algorithmic divergence.

5.1 Acquisition

1. Start the sensor after the standard 10 min warm-up and launch the frozen driver configuration.
2. Record one continuous static sequence for 60 min. A 30 min sequence is the minimum acceptable duration when laboratory time is constrained.
3. Record the point cloud, IMU, $/tf$ and $/tf_static$ streams. Record host reception time in addition to message timestamp when possible.
4. Sample process central processing unit use, memory use and network throughput every 1 s. Record sensor temperature if the driver exposes it.

- After the long run, perform five controlled stop-start cycles. Each cycle consists of a complete driver shutdown, a 10 s wait, relaunch and a 2 min recording. This secondary test measures restart reproducibility.

5.2 Metrics and failure definitions

Outcome	Definition	Required report
Point-cloud rate	Inverse of mean inter-arrival time; also median and 5th-to-95th percentile	Report messages per second and valid points per second
Timing jitter	Standard deviation, median absolute deviation and 95th percentile of inter-arrival intervals	Plot against time; identify periodic spikes
Timestamp integrity	Negative, zero or abnormally large timestamp increments	Any backward step is a critical failure
Dropout	Inter-arrival interval greater than five times the median interval	Report count, longest gap and total lost time
Invalid return fraction	Non-finite or physically invalid points divided by received points	Report overall and as a time series
IMU integrity	Rate, gaps, saturation and non-finite values	Compare measured rate with the stated output
Availability	Uninterrupted valid-stream duration divided by planned duration	Report restarts and human interventions
Resources	Mean, 95th percentile and temporal drift of CPU and memory	Separate driver, container and monitor

CRITICAL FAILURE: A timestamp reversal, driver crash, unrecoverable topic loss or undocumented restart invalidates the claim of uninterrupted stability. The raw recording remains part of the dataset and the failure is reported.

6. Experiment 4 - trajectory and map validation

This experiment links sensor behaviour to task-level odometry and mapping. It requires a reference hierarchy because map repeatability alone does not establish trajectory accuracy.

6.1 Reference hierarchy

- Preferred reference: calibrated motion-capture system, robotic total station or equivalent external pose measurement with documented uncertainty.
- Acceptable independent reference: a rigidly co-mounted, higher-density sensor such as an Ouster-64 processed with a validated reference pipeline. This is called an independent reference system, not absolute ground truth.
- Fallback: surveyed start, end and intermediate checkpoints combined with repeated-map comparison. This supports repeatability and endpoint error only.

The highest available level is selected before acquisition and stated in the paper. Claims are limited to that level.

6.2 Sensor mounting and calibration

When a reference sensor is used, both sensors are mounted on one rigid structure. The six-degree-of-freedom extrinsic transformation is calibrated and independently checked. Timestamp alignment is quantified. A synchronization error study is performed by shifting one stream by +/- 5, 10 and 20 ms during analysis to determine sensitivity.

6.3 Trajectory design

Use a 20 to 40 m indoor loop containing a straight corridor, at least two 90 degree turns, doors or recesses, and a return to the start. Mark the start pose and three checkpoints. Keep the environment static for the primary condition. Test two speeds when feasible: 0.20 m/s and 0.40 m/s. Perform five traversals per speed; three is the minimum. Parameters remain unchanged across all repetitions.

6.4 Processing and metrics

Use one fixed LiDAR-inertial odometry pipeline for the L1 and record its complete version and parameters. When a reference trajectory is available, calculate the Absolute Trajectory Error (ATE) after one stated rigid alignment and the Relative Pose Error (RPE) over fixed path-length intervals. Report translational and

rotational components. Also report loop-closure error, map-to-reference cloud distance, wall thickness and run success.

- ATE: root-mean-square position difference over the complete time-aligned trajectory after the declared alignment.
- RPE: relative motion disagreement over 1 m and 5 m path intervals.
- Loop-closure error: distance and orientation between estimated final and initial poses before any forced closure.
- Map distance: median and 95th percentile point-to-surface distance in static, planar regions after the stated alignment.
- Wall thickness: transverse dispersion across pre-selected wall segments, excluding doors, corners and dynamic objects.
- Success: completion without reset, localization divergence or loss of the required output stream. Divergence is defined before analysis.

If only the fallback reference is available, ATE and RPE are omitted. The paper reports checkpoint errors, loop closure and inter-run map consistency instead.

7. Statistical analysis plan

Analysis is performed at the acquisition or traversal level. Millions of points in one run do not increase the number of independent replicates. Raw per-run outcomes are retained and displayed where space permits.

7.1 Descriptive statistics and uncertainty

For each condition, report the mean, standard deviation, median, minimum, maximum and a 95% confidence interval. Use a non-parametric bootstrap over independent acquisitions with at least 10,000 resamples when the distribution is not adequately represented by a normal approximation. Fix and report the random seed.

7.2 Factor effects

For the range experiment, estimate the effect of distance and incidence angle using acquisition-level regression. A robust regression is preferred when residuals contain isolated extreme values. Material is analysed as a categorical factor. Report coefficients, confidence intervals and residual diagnostics. Inferential p-values are secondary to effect sizes and uncertainty. If multiple formal tests are performed, control the family-wise error rate with the Holm procedure.

7.3 Pre-defined exclusions and sensitivity analyses

No run is removed solely because its result is unfavourable. Repeat the principal calculations with and without protocol-invalid runs, while keeping both sets identifiable. Coverage is repeated at 1 degree and 2 degree angular resolution. Plane fitting is repeated with at least two reasonable RANSAC thresholds. Trajectory metrics are repeated with the declared time-alignment sensitivity shifts.

7.4 Claim discipline

- State manufacturer values as declared specifications, not observed results.
- Use the term accuracy only when a traceable or independently validated reference is available.
- Use repeatability for disagreement among repeated runs without external truth.
- Do not infer statistical independence from individual point counts.
- Report negative and failed runs with the same traceability as successful runs.

8. Data management and reproducibility

Raw data are immutable. Every derived file is reproducible from raw data and version-controlled code. Each recording has one manifest row and one cryptographic checksum. No figure is generated manually from copied values.

8.1 Directory structure

```

unitree_l1_validation/
  README.md
  metadata/experiment_manifest.csv
  metadata/reference_instruments.csv
  config/driver.yaml
  config/tf_static.yaml
  config/software_versions.txt
  bags/raw/
  logs/system/
  analysis/
  results/tables/
  results/figures/
  paper/

```

8.2 File naming and checksums

Use the pattern YYYYMMDD_experiment_condition_runNN. Example: 20260807_E1_distance_2m_run03. Generate a SHA-256 checksum immediately after closing each raw recording. The manifest stores the checksum, file size and acquisition duration.

8.3 Minimum metadata

Category	Required fields
Identity	File name, date and time, operator, sensor model, serial number, firmware
Geometry	Reference distance, incidence angle, material, target dimensions, sensor orientation, origin offset
Reference	Instrument model, serial number, calibration date, resolution, stated uncertainty
Software	ROS 2 distribution, driver commit, container digest, parameters, static transforms, analysis commit
Acquisition	Topics, duration, run number, warm-up, environment, nominal speed
Quality	Bag reopened, missing topics, excluded status, exclusion reason, anomalies
Traceability	SHA-256 checksum, raw size, analysis version, figure version

9. Quality assurance, safety and governance

A complete pilot run is processed end to end before the definitive matrix begins. This pilot verifies topic names, transforms, geometry, file naming, analysis scripts and reference calculations. Definitive acquisition starts only after the pilot produces a valid result and can be reproduced from raw data.

9.1 Go/no-go criteria

- Sensor model and origin are identified; optical window is clean.
- Mounting is rigid and the coordinate transform is fixed.
- Reference instrument and origin-offset method are documented.
- Point-cloud and IMU timestamps are monotonic in the pilot bag.
- Required topics are present and the bag reopens without corruption.
- One pilot dataset passes the complete processing pipeline.
- Raw-data storage and backup capacity are sufficient.

9.2 Safety

Trip hazards from cables, target supports and floor markers are controlled. The mobile-robot experiment uses a low speed, a clear emergency stop and a designated observer. The robot path is closed to uninvolved people during primary runs. The LiDAR is operated within manufacturer instructions. No participant data are collected; nevertheless, recordings containing identifiable people are not released without institutional approval.

9.3 Data ownership and authorship

Before public release, confirm data ownership, code licensing, author order and repository location with Prof. Urbano J. Nunes and Dr Tiago Barros. The protocol does not itself authorize publication of laboratory data.

10. Required outputs and decision rules

The protocol is complete only when each mandatory experiment has raw data, metadata, quality-control logs, analysis code and a concise result. A visually attractive map is not a substitute for uncertainty, repetitions or traceability.

10.1 Minimum figures

- Range signed error versus reference distance, with acquisition-level points and 95% confidence intervals.
- Point-to-plane dispersion and valid-return fraction versus incidence angle and material.
- Angular coverage versus accumulation time, accompanied by one grayscale angular hit-density map.
- Point-cloud and IMU timing over the complete stability run, including identified gaps.
- Trajectory and map comparison against the selected reference level, or inter-run repeatability when no independent reference exists.

10.2 Minimum tables

Only three result tables are required: sensor and software configuration; acquisition-level range results summarized by condition; and stream-stability outcomes. Trajectory results may be presented as a compact table when several speeds or algorithms are compared. Additional tables are omitted unless they improve interpretation.

10.3 Final scientific decisions

- H1 is supported only for conditions whose signed error and uncertainty are compatible with the specified threshold. Other conditions are reported separately.
- H2 is supported when coverage increases reproducibly with time and directional non-uniformity is quantified across independent runs.
- H3 is supported only when the planned duration completes without critical failures and timing metrics remain reported, not merely visually acceptable.
- H4 is supported at the accuracy level only with an independent validated reference. Otherwise, the conclusion is limited to repeatability and geometric consistency.

Appendix A. Acquisition checklist

- ☐ Sensor variant, serial number and firmware recorded
- ☐ Optical window clean; standard warm-up completed
- ☐ Mount, orientation and transforms unchanged
- ☐ Reference distance, angle and origin offset measured
- ☐ Bag name and manifest row created before recording
- ☐ Point-cloud, IMU and transform topics present
- ☐ Environment static; no person in the measurement region
- ☐ Bag reopened; duration, message counts and timestamps checked
- ☐ Anomalies and protocol deviations recorded immediately
- ☐ SHA-256 checksum generated; backup completed

Run decision: ☐ VALID ☐ INVALID BY PRE-DEFINED RULE ☐ REPEAT REQUIRED

Reason or deviation:

Operator initials: _____ Time: _____

Appendix B. Experiment manifest template

Field	Definition	Example
bag_name	Unique name following the naming convention	20260807_E1_distance_2m_run03
experiment_id	E1, E2, E3 or E4	E1
condition	Distance, angle, material, coverage pose, stability or speed	distance_2m
run_number	Independent acquisition index	3

Field	Definition	Example
date_time_local	Start time with time zone	2026-08-07T10:14:00+01:00
sensor_model_serial	Exact model and serial number	L1 PM / serial ...
reference_value	Distance, angle or reference-system identifier	2.000 m
duration_s	Actual recording duration	20.0
software_commit	Driver and analysis commit	git hash
validity	valid / invalid / repeat	valid
exclusion_reason	Pre-defined reason when invalid	
sha256	Checksum of raw file	64-character value
notes	Observed anomaly or environmental condition	No motion observed

References

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- [2] Unitree Robotics, Unitree 4D LiDAR-L1 User Manual, version 1.1, June 2024.
- [3] B. Muralikrishnan, "Performance Evaluation of Terrestrial Laser Scanners - A Review," Measurement Science and Technology, vol. 32, no. 7, article 072001, 2021, doi: 10.1088/1361-6501/abdae3.
- [4] S. Soudarissanane, R. Lindenbergh, M. Menenti, and P. J. G. Teunissen, "Incidence Angle Influence on the Quality of Terrestrial Laser Scanning Points," International Society for Photogrammetry and Remote Sensing Workshop Laser Scanning, 2009.
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- [8] Z. Zhang and D. Scaramuzza, "A Tutorial on Quantitative Trajectory Evaluation for Visual(-Inertial) Odometry," IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 7244-7251, 2018, doi: 10.1109/IROS.2018.8593941.

PROTOCOL COMPLETION CRITERION: The final dataset must allow another researcher to reconstruct every reported number from immutable raw recordings, documented references, the experiment manifest and version-controlled analysis code.